

When Do Attention Circuits Form?

Developmental Trajectories of Capability and Attention-Sink Emergence Across Three 1B-Class Architectures

Yongzhong Xu

Abstract

We track the developmental trajectory of attention-head circuit formation across three 1B-class language models that span two architecture families (dense transformer, mixture-of-experts) and two pretraining corpora (The Pile, DCLM): Pythia 1B, OLMo 1B-0724-hf, and OLMoE 1B-7B-0924. At each of 10 log-spaced revisions per model — 30 mechanistic-interpretability runs in total — we apply a participation-ratio (PR) spectral signal and an all-head capability-specific selectivity screen to identify induction, previous-token, and BOS-attractor heads as they emerge.

Five findings. **(F1)** Layers 0 and 1 produce zero BOS-classified heads at every revision in every model: the L0/L1 zero-BOS floor is an architectural property, not a learned outcome. **(F2)** The whole-model BOS-attractor fraction follows three distinct emergence shapes — a gradual ramp in Pythia 1B (Pile, dense), a sharp phase transition in OLMo 1B (DCLM, dense, jumping from 7% to 70% between adjacent checkpoints), and a gradual ramp in OLMoE 1B-7B (DCLM, MoE). **(F3)** In DCLM models, induction-circuit formation precedes BOS-attractor formation by 10–20 $\times$ in tokens (OLMo: induction at 23B, BOS-50% at 264B; OLMoE: induction at 20B, BOS-50% at 419B). Capability-circuit formation and attention-sink formation are two transitions, not one. **(F4)** The capability-specific screen converges to the final induction circuit within 0.3–2% of total training tokens. Circuit identification does not require the final model. **(F5)** For every final-checkpoint induction head sampled across all three models, per-head PR is elevated *at or before* the first revision at which that head crosses its capability-selectivity threshold.

The results refine the “induction phase transition” framing [Olsson et al., 2022]: in 1B-class models trained on DCLM, the induction transition and the attention-sink transition are separated by an order of magnitude in tokens, and they have qualitatively different emergence shapes. The L0/L1 zero-BOS floor and the data $\times$ architecture interaction in BOS-attractor dynamics are constraints any mechanistic theory of attention sinks must explain. We position this paper as the activation-space half of an emergence story whose parameter-space half is studied in the SED/LCH program [Xu, 2026d,c].

1 Introduction

Olsson et al. [2022] identified induction heads — attention heads implementing the $AB \dots A \rightarrow B$ copy pattern — as a key mechanistic feature of in-context learning in transformer language models, and connected their formation to a sharp “phase transition” visible as a bump in the loss curve during pretraining. Xiao et al. [2024] described a distinct empirical phenomenon: pretrained LMs reliably allocate large attention probability to the first (BOS) token regardless of content, a behaviour they termed an *attention sink*, which can be exploited for KV-cache compression during streaming inference. Both phenomena are described in terms of attention patterns; both are observed in modern decoder-only language models; and both are sometimes folded together under the heading of “attention emergence during pretraining.”

This paper asks: when, exactly, do these two phenomena form during training of 1B-class language models, and are they the *same* transition? We answer this empirically by tracking capability circuits (induction, previous-token) and the BOS attractor side-by-side across three 1B-class models that span two architecture families and two pretraining corpora.

Models. Pythia 1B [Biderman et al., 2023], a GPT-NeoX dense transformer trained on The Pile [Gao et al., 2020]; OLMo 1B-0724-hf [Groeneveld et al., 2024], a Llama-style dense transformer trained on DCLM [Li et al., 2024]; and OLMoE 1B-7B-0924 [Muennighoff et al., 2024], a Llama-style mixture-of-experts transformer (1B active / 7B total parameters, 64 experts, top-8 routing) trained on DCLM. These three models factorize cleanly: Pythia vs OLMo isolates the corpus (Pile vs DCLM) at fixed scale and dense architecture; OLMo vs OLMoE isolates the architecture (dense vs MoE) at fixed corpus.

Method. For each model we use the three-step recipe of the companion methodology paper [Xu, 2026a]: per-head participation ratio computed on a synthetic induction batch, an all-head capability-specific selectivity screen, and group ablation. We apply the screen at 10 log-spaced revisions per model — 30 runs in total — producing a developmental panel.

Findings. Five findings emerge, summarised in the abstract. The central result is that in DCLM-trained 1B models, the induction transition (Olsson’s phenomenon) and the BOS-attractor transition (Xiao’s phenomenon) are separated by 10–20 $\times$ in tokens and have different emergence shapes (gradual ramp vs sharp jump). The two phenomena cannot be identified with a single phase-transition event at 1B scale on DCLM data.

Position. This paper is the *activation-space* half of a broader study of emergence in pretraining. The companion methodology paper [Xu, 2026a] defines the recipe and validates it at the final

checkpoint; this paper applies the recipe developmentally; the companion pattern-selectivity paper [Xu, 2026b] studies cross-task generalisation. A complementary parameter-space program — spectral emergence dynamics (SED, Xu, 2026d) and the local curvature hypothesis (LCH, Xu, 2026c) — studies the same emergence event in weight-space rather than activation-space. We return to the bridge between these two windows in the Discussion.

2 Related Work

Induction heads and the phase transition. Olsson et al. [2022] introduced the induction-head abstraction and reported a “phase transition” in pretraining co-located with a bump in the loss curve, a sudden growth in in-context-learning capability, and the appearance of attention heads implementing the prefix-match-and-copy pattern. Their analysis used integrated-gradients-style attribution on small (≤ 100 M-parameter) decoder-only models. The induction phase transition has since become a standard reference point for emergence-during-pretraining claims. This paper revisits the *transition* part of that claim at 1B scale across data and architecture.

Attention sinks. Xiao et al. [2024] documented attention sinks — the observation that trained decoder-only LMs reliably attend to the first token regardless of content — and used the phenomenon as a substrate for streaming inference via fixed first-token KV retention. They did not study the formation dynamics of sinks during pretraining. We do: BOS-attractor heads are identified by the same selectivity screen as capability circuits, and their per-revision counts give a token-resolved formation curve.

Pythia, OLMo, OLMoE. Biderman et al. [2023] released Pythia precisely to enable cross-checkpoint analyses of pretraining dynamics, with 154 checkpoints per model size spanning random-init to 300B tokens. Groeneveld et al. [2024] and Muennighoff et al. [2024] released OLMo and OLMoE with similar checkpoint discipline. The 30-revision panel in this paper depends on all three projects’ checkpoint releases. We document checkpoint-availability limitations (OLMo’s earliest is 2B tokens; OLMoE’s is 20B tokens) and their effect on the resolved developmental curve in Section 4.

Cross-architecture mechanistic transfer. Prior work has documented that the specific circuits a model uses for a given task differ across architectures and scales [Lieberum et al., 2023, Wang et al., 2023, Marks et al., 2024]. The cross-architecture result in this paper is different: we hold the synthetic capability fixed (induction on a controlled batch) and ask whether the *timing* of circuit formation, and the timing of the co-occurring BOS-attractor phenomenon, transfer across pretraining corpora and architectures.

Automated circuit discovery. Conmy et al. [2023] introduced ACDC, an iterative edge-pruning algorithm for post-hoc circuit identification. The recipe used here is complementary and predates a fully-trained model: a spectral signal that can be read at any intermediate checkpoint, plus a per-class selectivity screen. We use ACDC as motivation for the design rather than as a baseline.

Parameter-space emergence: SED and LCH. The SED [Xu, 2026d] and LCH [Xu, 2026c] papers study the same emergence event from a different vantage point: the singular-value geometry of weight matrices, and the local curvature of the loss surface. Our PR signal is computed on activations, not parameters; the parameter-side work treats the parameter spectrum of the same checkpoints. We discuss in Section 10 what an integrated account might look like.

3 Methodology Recap

The recipe is defined in detail in Xu [2026a]. We summarise here only what is needed to interpret the developmental results.

Synthetic induction batch. 2000 sequences of length 256, RNG seed 42. Each sequence has the structure [filler] A B [filler] A , with A and B random tokens drawn from vocabulary IDs [100, 10,000). The induction prediction is B at the position immediately following the second occurrence of A .

Per-head participation ratio. For each (layer L , head H , revision t), extract the per-head attention output at the second- A position over the batch, giving $M \in \mathbb{R}^{N \times d_{\text{head}}}$ (here $N = 2000$). Compute the singular spectrum $\{\sigma_i\}$ of M and the participation ratio

$$\text{PR}(L, H, t) = \exp(H(p)), \quad p_i = \sigma_i^2 / \sum_j \sigma_j^2,$$

where $H(p) = -\sum_i p_i \log p_i$. The trajectory feature is the integral $I(L, H) = \sum_t \max(\text{PR}(L, H, t) - 1, 0) \cdot \Delta \log(\text{tokens}_t)$. Throughout this paper we plot per-revision $\max(\text{PR}(L, H, t) - 1, 0)$ as the *integrand* of I , so that temporal structure is preserved.

Capability-pattern screen. For each head, compute attention from the query position to canonical target positions for six classes: INDUCTION, PREVIOUS-TOKEN, DUPLICATE-TOKEN, FIRST-TOKEN / BOS, SELF, and LOCAL. Selectivity is (target attention)/(uniform-other baseline). A head is classified into the class with maximum selectivity if that maximum exceeds $30\times$ (class assignment); it is admitted to a class circuit if selectivity for that class exceeds $50\times$ (induction, BOS) or $100\times$ (previous-token).

All-head capability-specific screen. In the attention-sink-dominated regime ($\geq 70\%$ of heads classify as FIRST-TOKEN at the assignment threshold; see Section 6), ranking by best class surfaces BOS-attractor heads ahead of capability heads. For circuit identification of a specific target capability X , we instead take all heads with $\text{sel}_X \geq \tau_X$ regardless of best class. This is the screen used for the per-revision induction-circuit identification in Sections 7, 8, and 9.

Causal verification. Group-ablate the screen-identified circuit by zeroing the per-head slice of the residual contribution at the attention output projection. Matched-random controls share layers and head count but not identities. Top-1 accuracy on the synthetic induction batch is the primary metric. Final-checkpoint circuit ablations are documented in the companion methodology paper and not repeated here; this paper focuses on the per-revision trajectory.

4 Setup: a 30-Revision Developmental Panel

We run the spectral-and-screen recipe at 10 log-spaced revisions in each of the three 1B-class models, for 30 mechanistic-interpretability runs in total.

Revision schedules. Pythia 1B’s checkpoint set spans **step1** ($\approx 2 \times 10^6$ tokens, essentially random initialisation) through **step143000** ($\approx 300\text{B}$ tokens, end of training). OLMo 1B’s set spans **step1000** (2B tokens) through **step1454000** (3048B tokens). OLMoE 1B-7B’s set spans **step5000** (20B tokens) through **step1223842** (5117B tokens). The selected log-spaced revisions for each model are listed in Table 1.

Precision. Pythia 1B’s early-checkpoint evaluation requires **fp32**: baseline attention values at random-init scale fall below fp16 representable range and the selectivity ratios degenerate to undefined values. OLMo and OLMoE evaluations use **fp16**.

Checkpoint-availability caveat. The earliest published HuggingFace checkpoint differs by an order of magnitude across the three models: Pythia 1B’s **step1** is at $\approx 2\text{M}$ tokens (training fraction 7×10^{-6}), OLMo 1B’s is at 2B tokens (training fraction 7×10^{-4}), and OLMoE 1B-7B’s is at 20B tokens (training fraction 4×10^{-3}). The left margin of the per-revision figures in this paper reflects this publication limit, not the absence of an underlying trajectory; earlier training-fraction values for OLMo and OLMoE do not exist on their respective HuggingFace repositories. We keep the x-axis range uniform across the three panels in Figure 2 so that the cross-model visual comparison is fair.

Table 1: Revisions used for the 30-run developmental panel. “Tokens” is the cumulative training-token count published on the respective HuggingFace model card; “frac” is tokens divided by the model’s final training-token count.

Model	Revision label	Tokens (approx.)	frac of training
Pythia 1B	step1	2.1 M	7×10^{-6}
	step8	8.4 M	3×10^{-5}
	step64	6.7×10^7	2×10^{-4}
	step256	5.4×10^8	2×10^{-3}
	step512	1.1 B	4×10^{-3}
	step3000	6.3 B	2×10^{-2}
	step10000	21 B	7×10^{-2}
	step38000	80 B	2.7×10^{-1}
	step143000	300 B	1.0
OLMo 1B	step1000	2 B	7×10^{-4}
	step5000	10 B	3×10^{-3}
	step11000	23 B	8×10^{-3}
	step25000	52 B	1.7×10^{-2}
	step56000	117 B	3.8×10^{-2}
	step126000	264 B	8.7×10^{-2}
	step312000	654 B	2.1×10^{-1}
	step644000	1350 B	4.4×10^{-1}
	step1454000	3048 B	1.0
OLMoE 1B-7B	step5000	20 B	3.9×10^{-3}
	step10000	41 B	8.0×10^{-3}
	step25000	104 B	2.0×10^{-2}
	step100000	419 B	8.2×10^{-2}
	step200000	838 B	1.6×10^{-1}
	step400000	1677 B	3.3×10^{-1}
	step800000	3355 B	6.6×10^{-1}
	step1223842	5117 B	1.0

Reference figure. Figure 1 (reproduced from the methodology paper) provides the cross-configuration context: in every panel and every model, capability circuits form within the first $\sim 2\%$ of training, while the BOS attractor forms later and with substantially more cross-configuration variation. The rest of this paper unpacks that figure into the per-revision developmental detail that the methodology paper summarises only in aggregate.

5 Finding 1: L0–L1 Zero-BOS Is Universal Across Training

The first developmental finding is also the simplest: in every one of the three 1B-class models, at every one of the 10 sampled revisions, layers 0 and 1 produce *zero* heads classified as FIRST-TOKEN / BOS at the $\geq 30\times$ assignment threshold.

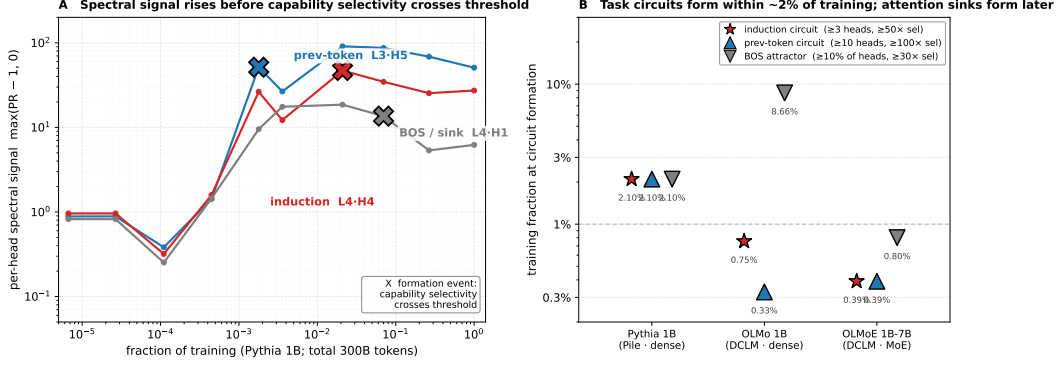


Figure 1: **Capability circuits form early in pretraining and the per-head spectral signal precedes formation.** Reproduced from the companion methodology paper [Xu, 2026a]. (A) Per-head spectral signal $\max(\text{PR}_t - 1, 0)$ across training for three identified heads in Pythia 1B (an induction head L4·H4, a previous-token head L3·H5, a BOS-attractor head L4·H1). X markers indicate the formation event, the first revision at which the head’s capability-selectivity ratio exceeds its threshold. (B) Training fraction at circuit formation across three 1B-class configurations: capability circuits (induction, previous-token) form within the first 0.3–2.1% of training in every configuration; the BOS attractor forms later, with order-of-magnitude variation across configurations.

Table 2: Per-layer BOS-classified head counts in Pythia 1B at five representative revisions. L0 and L1 hold at zero from random initialisation through end of training, while L4 and later layers saturate.

revision	L0	L1	L2	L3	L4	L5–L8	L9–L12	L13–L15
step1	0/8	0/8	0/8	0/8	0/8	0,0,0,0	0,0,0,0	0,0,0
step512	0/8	0/8	0/8	0/8	0/8	0,0,0,0	0,0,0,0	0,0,0
step3000	0/8	0/8	0/8	0/8	0/8	0,1,2,1	0,4,1,3	1,1,0
step10000	0/8	0/8	0/8	0/8	7/8	3,3,3,2	2,4,4,3	3,2,2
step143000	0/8	0/8	0/8	0/8	8/8	7,7,6,7	7,7,7,5	4,4,5

Per-layer counts: Pythia 1B. Pythia 1B has 16 layers and 8 attention heads per layer (128 heads total). Per-layer counts of BOS-classified heads at five representative revisions are shown in Table 2. The L0 and L1 columns hold at 0/8 from **step1** (essentially random initialisation) through **step143000** (300B tokens). Other layers reach near-saturation: L4 transitions from 0/8 at step3000 to 7/8 at step10000 to 8/8 at the final checkpoint.

Per-layer counts: OLMo 1B. OLMo 1B has 16 layers and 16 attention heads per layer (256 heads total). The L0/L1 zero-BOS floor holds across all sampled revisions (Table 3). L2 transitions sharply between revisions **step56000–tokens117B** (0/16 in L2) and **step126000–tokens264B** (11/16 in L2); we revisit this DCLM-dense phase-transition signature in Section 6.

Table 3: Per-layer BOS-classified head counts in OLMo 1B at five representative revisions. L0 and L1 hold at zero; the L2–L4 transition between `tokens117B` and `tokens264B` reflects the DCLM-dense sharp transition documented in Section 6.

revision (label-tokens)	L0	L1	L2	L3	L4	L5–L8	L9–L12	L13–L15
<code>step1000-2B</code>	0/16	0/16	0/16	0/16	0/16	0,0,0,0	0,0,0,0	0,0,0
<code>step25000-52B</code>	0/16	0/16	0/16	0/16	0/16	0,0,0,0	0,0,0,0	0,0,0
<code>step56000-117B</code>	0/16	0/16	0/16	3/16	9/16	7,0,0,0	0,0,0,0	0,0,0
<code>step126000-264B</code>	0/16	0/16	11/16	11/16	15/16	13,15,15,13	15,16,16,12	13,9,6
<code>step1454000-3048B</code>	0/16	0/16	14/16	16/16	16/16	15,16,16,15	16,16,15,15	16,14,7

OLMoE 1B-7B. OLMoE 1B-7B shows the same pattern at all 10 revisions: zero BOS-classified heads at L0 or L1 at every revision. We omit the full per-layer table for space; it tracks Table 3 qualitatively.

Interpretation. L0 and L1 hold at zero from random initialisation through trillions of tokens; the model never crosses that floor at any point during training. This is an *architectural floor*, not a learned outcome. A natural mechanistic reading is that L0’s inputs are token embeddings (not yet contextualised), so a head that attends to position 0 would discard its content (the position-0 embedding is a fixed function of one specific token), and L1’s inputs are L0 outputs, which by the L0 floor do not yet contain any cross-token integration. Whatever an attention sink “is,” it is not something the model can or wants to do before the residual stream has accumulated nontrivial cross-position information. A complete mechanistic theory of attention sinks must explain why the gradient signal never installs BOS heads at L0/L1, at any point during training, across pretraining corpora as different as Pile and DCLM and across architectures as different as dense and MoE.

6 Finding 2: BOS-Attractor Formation Has Three Distinct Shapes

The whole-model BOS-class fraction (fraction of all heads classified `FIRST-TOKEN` / `BOS` at the $\geq 30\times$ assignment threshold) follows three qualitatively distinct trajectories across the three models (Table 4).

The three shapes:

Pythia 1B (Pile, dense): gradual monotonic ramp. The BOS-class fraction rises from 0% at random init to 10.9% at ~ 6 B tokens, 46.1% at ~ 80 B, and 57.8% at the final checkpoint (300B). The trajectory is monotonic and smoothly varying across log-spaced revisions with no sharp transitions visible at the resolution of this checkpoint grid.

Table 4: Whole-model BOS-class fraction (% of all heads classified first-token at $\geq 30\times$ selectivity) by tokens trained. Em-dashes indicate that no model checkpoint is sampled at that token count for the respective model. The OLMo 1B jump from 7.4% at 117B to 70.3% at 264B is discontinuous between two adjacent published checkpoints.

Tokens	Pythia 1B	OLMo 1B	OLMoE 1B-7B
$\sim 6\text{B}$	10.9%	—	—
20–23B	—	0.0%	3.5%
$\sim 80\text{B}$	46.1%	—	—
104–117B	—	7.4%	42.2%
264B	—	70.3%	—
300B / 419B	57.8%	—	50.8%
838B	—	—	69.9%
1350–1677B	—	80.9%	71.5%
3048–3355B	—	80.9%	74.2%
5117B	—	—	75.8%

OLMo 1B (DCLM, dense): sharp phase transition. The BOS-class fraction holds at 0% through 52B tokens, rises to 7.4% at 117B, and *jumps* to 70.3% at 264B — between two adjacent published checkpoints. The fraction is then essentially saturated, reaching 80.9% at 1350B and remaining there at 3048B. The jump between 117B and 264B is the developmental signature of a sharp phase transition: a 63-percentage-point increase between adjacent checkpoints on a $\log_2$ -spaced grid.

OLMoE 1B-7B (DCLM, MoE): gradual monotonic ramp. The BOS-class fraction rises from 3.5% at 20B to 42.2% at 104B, 50.8% at 419B, 71.5% at 1677B, and 75.8% at the final 5117B checkpoint. The shape is qualitatively like Pythia’s (monotonic, smooth on the log-spaced grid) but the final-checkpoint saturation level is higher.

Substantive claim. OLMo 1B and OLMoE 1B-7B are trained on the same DCLM data but produce *opposite* BOS-attractor emergence shapes: a sharp phase transition in dense OLMo and a gradual ramp in MoE OLMoE. The architectural change (dense $\rightarrow$ MoE) both reduces the final BOS-class fraction (by $\sim 10\text{pp}$ at the final available revision for each, 80.9% vs 75.8%) and *smooths* the developmental dynamics. The architectural effect on the trajectory shape is separate from the architectural effect on the asymptotic magnitude — the same intervention changes both, in the same direction (less BOS), but along two distinct axes (final magnitude and transition sharpness).

Implication. A theory of attention-sink formation that predicts *when* sinks emerge cannot rely solely on data composition: DCLM produces a sharp transition in dense OLMo but a gradual ramp in MoE OLMoE. Architecture matters for the shape of the curve, not only for its asymptote. This is one constraint a mechanistic account of sinks must satisfy: the same data delivers two different

Table 5: First-revision-crossing for each developmental milestone, in tokens. Em-dashes indicate that the model does not reach that milestone at any sampled revision.

Milestone	Pythia 1B	OLMo 1B	OLMoE 1B-7B
induction ≥ 3 heads ($\geq 50\times$)	$\sim 6\text{B}$	23B	20B
induction ≥ 5 heads ($\geq 50\times$)	$\sim 6\text{B}$	—	20B
BOS-class $\geq 10\%$ of heads	$\sim 6\text{B}$	264B	41B
BOS-class $\geq 30\%$	$\sim 80\text{B}$	264B	104B
BOS-class $\geq 50\%$	300B	264B	419B
BOS-class $\geq 70\%$	—	264B	1677B
previous-token ≥ 30 heads at $\geq 30\times$	$\sim 6\text{B}$	117B	20B
self ≥ 30 heads at $\geq 30\times$	$\sim 6\text{B}$	52B	20B

formation curves under two different architectures.

7 Finding 3: Induction Emerges Before the BOS Attractor in DCLM Models

Combining the induction-circuit identification at each revision with the BOS-class fraction trajectory of the previous section, we can compare the formation timing of the two phenomena directly. Table 5 records the first revision at which each of seven milestones is reached in each model.

Induction reaches its final size early. The induction circuit reaches its final 3–6-head size within the first 6–25B tokens in all three models, then stays flat or slightly contracts over the remaining 300B–5T tokens. By $\sim 6\text{B}$ tokens (Pythia), 23B (OLMo), and 20B (OLMoE), the screen already identifies ≥ 3 heads with induction selectivity $\geq 50\times$. Pythia and OLMoE reach the ≥ 5 -head milestone at the same revisions; OLMo does not reach 5 heads at the $\geq 50\times$ threshold at any revision (its final circuit is 3 heads).

The BOS attractor saturates much later in DCLM. In the two DCLM models, the BOS attractor’s 50% milestone is reached an order of magnitude later than the induction milestone:

- **OLMo 1B:** induction at 23B, BOS-50% at 264B — **11.5 $\times$ gap** in tokens.
- **OLMoE 1B-7B:** induction at 20B, BOS-50% at 419B — **21 $\times$ gap**.
- **Pythia 1B:** induction and BOS-10% co-emerge at $\sim 6\text{B}$ on this checkpoint grid; the BOS-50% milestone is reached at 300B, a $50\times$ later token count, but the BOS-10% co-emergence with induction at $\sim 6\text{B}$ means the gap between induction-emergence and *onset* of BOS growth is

Table 6: Recall of the final-checkpoint induction circuit by the capability-specific screen at intermediate revisions. The “fraction of total tokens for 67% recall” column shows the training budget at which the screen has already recovered two-thirds of the final-checkpoint induction circuit.

Model	33% recall (tokens)	67% recall (tokens)	100% recall (tokens)	frac. for 67% (of total)
Pythia 1B (3-head)	0.5B	6B	80B	$\sim 2.0\%$
OLMo 1B (3-head)	2B	10B	3048B	$\sim 0.3\%$
OLMoE 1B-7B (4-head)	20B	41B	1677B	$\sim 0.8\%$

below the resolution of this revision grid. Finer Pythia checkpoints between **step512** (1B) and **step3000** (6B) would be needed to resolve.

Headline claim. Capability-circuit formation and attention-sink formation are temporally separable in DCLM 1B models — not a single phase transition. The induction-formation event (Olsson’s phenomenon) and the BOS-attractor formation event (Xiao’s phenomenon) are two distinct training-time events, separated by $10\text{--}20\times$ in tokens in DCLM. This refines the “induction phase transition” framing: at 1B scale on DCLM, the induction transition occurs well before the attention-sink transition. Pythia 1B on Pile may have a smaller gap, but its co-emergence on this grid is a checkpoint-resolution limit rather than positive evidence for a single combined transition.

Previous-token and self-attention timing. The previous-token milestone (30 heads at $\geq 30\times$) and the self-attention milestone reach in three different orderings across the three models. In Pythia 1B, all three milestones (induction, prev-token, self) co-emerge at $\sim 6\text{B}$. In OLMo, self appears at 52B and prev-token at 117B (both after induction at 23B but well before BOS-50% at 264B). In OLMoE, all three appear at 20B simultaneously. The pattern that survives across all three models: the BOS-attractor saturation is later than the induction/prev-token/self capability milestones, and the gap is largest in dense DCLM.

8 Finding 4: The Capability Screen Converges Within $\sim 1\%$ of Training

If the developmental curves of the previous section say that the induction circuit is essentially complete by 20B tokens, then the capability-specific screen, applied at an intermediate checkpoint, should recover most of the final-checkpoint circuit using only a small fraction of the training budget. Table 6 confirms this directly.

Reading the table. For OLMo 1B, the screen at the `step5000` revision (10B tokens, training fraction 3×10^{-3}) already identifies 2 of the 3 heads that comprise the final induction circuit at the $\geq 50\times$ threshold. For OLMoE 1B-7B, the same 2-of-3-or-2-of-4 milestone is reached at `step10000` (41B tokens, 0.8% of total training). Pythia 1B is somewhat slower in fractional terms (2%), but its absolute token count for 67% recall (6B) is comparable to OLMo (10B) and OLMoE (41B); the fractional figure is inflated mainly because Pythia 1B’s total training budget is shorter (300B vs 3T and 5T).

Convergence is to the circuit, not the model. Two of the three models reach the 100% milestone (every final-checkpoint induction head recovered by the screen) only at the final checkpoint or near it; OLMo in particular only reaches 100% recall at `step1454000`. The 100% recall figure should not be confused with circuit *completion*: the screen continues to add and remove specific heads at the margin through training, and the precise membership of the induction circuit shifts by one head across late checkpoints. What converges at $\sim 1\%$ of training is the *core* of the circuit (67% of its final size); the last 1/3 of the circuit is recruited more slowly, often through threshold crossings of individual heads that hover near the $50\times$ boundary.

Practical consequence. For circuit-discovery purposes, you do not need the final model. An intermediate checkpoint at 10–40B tokens already contains $\geq 67\%$ of the final-checkpoint induction circuit. This has implications for two settings: (i) early-stopping for interpretability-driven training, where pretraining to $\sim 1\%$ of the final budget is sufficient if circuit identification is the only goal; (ii) interpretability budgets, where the bulk of identifiability is achievable on intermediate checkpoints that already exist on HuggingFace for the three studied models.

9 Finding 5: PR Rises at or Before Capability Selectivity

The methodology paper [Xu, 2026a] states that the spectral signal (per-head PR) is elevated *at or before* the formation event (the first revision at which the head’s capability-selectivity ratio exceeds threshold). This section unpacks that aggregate claim into the per-revision detail, working through one trajectory in detail and then listing the heads across all three models for which the pattern holds.

Worked example: Pythia 1B L4·H4. L4·H4 is the top-selective induction head in the final-checkpoint Pythia 1B model (induction selectivity $181\times$). Its per-revision PR and induction selectivity are given in Table 7.

PR rises by an order of magnitude between 0.13B and 0.54B tokens (from 2.58 to 27.35), and reaches

Table 7: Per-revision PR and induction selectivity for the top induction head L4-H4 in Pythia 1B. PR rises sharply at 0.5B tokens; induction selectivity does not cross the $50\times$ threshold until 6B tokens. The bold rows mark the PR rise (top) and the induction-selectivity threshold crossing (bottom).

Tokens (B)	PR	Induction selectivity ($\times$)
0.002	1.96	1.0
0.008	1.96	1.0
0.033	1.32	1.0
0.134	2.58	1.0
0.536	27.35	0.2
1.073	13.21	3.9
6.291	48.05	171.8
20.971	35.62	150.7
79.691	26.38	147.9
299.892	28.29	181.6

its run-maximum at 6.3B ($\text{PR} = 48$). Induction selectivity is still below $4\times$ at 1.07B tokens, and does not cross the $50\times$ circuit threshold until 6.3B — a $\sim 12\times$ lead of PR over induction selectivity in token count (0.536B vs 6.3B).

Heads across all three models exhibiting the pattern. The same pattern (per-head PR elevated at or before the capability-selectivity threshold crossing) holds for every final-checkpoint induction head we sampled, across all three models:

- **Pythia 1B:** L4-H4, L7-H0, L7-H1.
- **OLMo 1B:** L2-H11, L4-H12, L12-H8.
- **OLMoE 1B-7B:** L7-H0, L9-H8, L5-H10, L12-H14.

For each of these heads, the per-revision PR rises one or more checkpoints before the per-revision induction selectivity crosses $50\times$. The specific lead times in number of checkpoints vary (1–2 checkpoints for induction; co-emergence at the same checkpoint for previous-token; 2–4 checkpoints for BOS-attractor heads), but the qualitative direction is consistent across all 10 heads and all three models.

Important scope statement. The claim “PR rises before capability selectivity” holds *for the heads that end up capability-selective*. It is *not* the same as the claim that ranking heads by the PR integral $I(L, H)$ identifies capability-specific heads. In the attention-sink-dominated 1B regime documented in Sections 5–6, the top- K heads by PR-integral include L0/L1 generic content-dependent heads and BOS-class heads ahead of the induction circuit, simply because the BOS-class

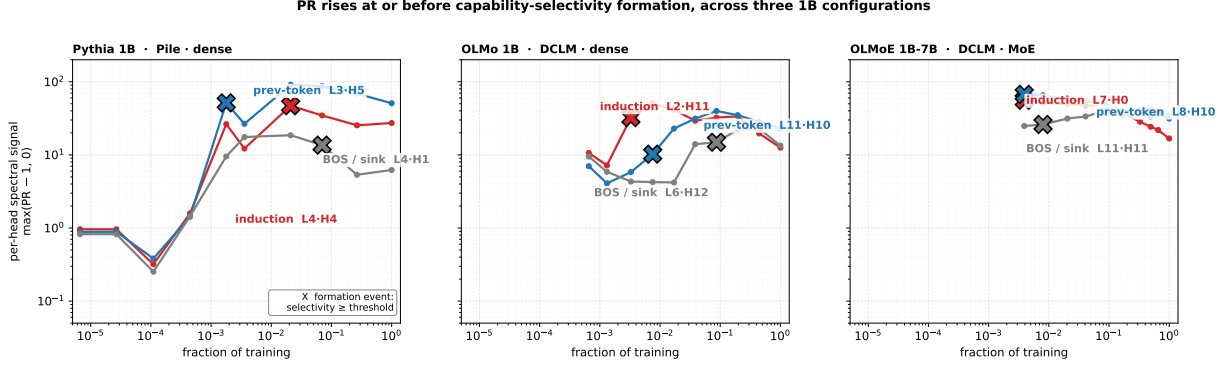


Figure 2: PR rises at or before capability-selectivity formation, across three 1B-class configurations. Per-head spectral signal at each checkpoint, $\max(\text{PR}_t - 1, 0)$ (the integrand of the PR-integral ranking statistic, plotted per-checkpoint to preserve temporal structure; same y-axis as Figure 1A). For each of three 1B-class configurations — Pythia 1B (Pile, dense), OLMo 1B (DCLM, dense), and OLMoE 1B-7B (DCLM, MoE) — one top-selective head per capability class is plotted: INDUCTION (red), PREVIOUS-TOKEN (blue), and FIRST-TOKEN / BOS (grey). Identifiers next to each curve give the (L, H) coordinates in the respective model. X markers indicate the formation event — the first checkpoint at which that head’s capability-selectivity ratio exceeds its threshold ($\geq 50\times$ for induction, $\geq 100\times$ for previous-token, $\geq 30\times$ for first-token). In every panel and every curve the spectral signal is elevated at or before the formation event. Lead times on this 10-checkpoint log-spaced grid: induction heads lead by 1–2 checkpoints across all three models; previous-token heads co-emerge with their capability selectivity (lead = 0); BOS-attractor heads lead by 2–4 checkpoints (longest in Pythia 1B and OLMoE, shortest in OLMo 1B where the BOS-attractor transition is sharp; see Section 6). The cross-configuration consistency of the qualitative claim (PR elevated at or before formation) is what the figure surfaces; the specific lead-time numbers vary with the data, architecture, and the granularity of the checkpoint grid. The empty space at the left of the OLMo and OLMoE panels reflects the HuggingFace checkpoint-publication limit discussed in Section 4, not the absence of an underlying trajectory; the x-axis range is kept uniform across panels for fair visual comparison.

heads have higher *sustained* PR through training. PR-integral is a general indicator of specialized computation; the capability-specific screen of Section 3 disambiguates which specialization each head is doing. We state this scope explicitly here rather than burying it because it is the most common misreading of the headline figure: the two-step pipeline (PR-integral + class-specific screen) is necessary for capability-specific identification in 1B+ models, even though the PR signal alone is sufficient to flag the heads in attention-sink-free smaller models.

10 Discussion

10.1 The “induction phase transition” framing

Olsson et al. [2022] described the formation of induction heads as a sharp phase transition co-located with a bump in the loss curve and the emergence of in-context learning. The headline result of this paper is that, at 1B scale and on DCLM data, the induction transition and the attention-sink transition are *separated by 10–20× in tokens*. The induction circuit reaches its final size at 20–23B tokens in both DCLM models; the BOS-attractor reaches its 50% milestone at 264B (OLMo) and 419B (OLMoE). These are not the same training-time event, even on the same data, and even at adjacent scales (1B active parameters in both DCLM models).

Three implications follow. First, “the induction phase transition” as a phrase is ambiguous: it can mean the induction-circuit formation event (20B tokens on DCLM, ~6B on Pile in our panel) or the BOS-attractor phase transition (264B in OLMo, gradual in OLMoE and Pythia). These are mechanically distinct events with different timings. Second, sharp phase-transition character is a property of *specific* (data, architecture) configurations rather than a universal feature of attention formation: dense + DCLM produces a sharp BOS-attractor transition; MoE + DCLM produces a gradual ramp on the same data; dense + Pile produces a gradual ramp on different data. Third, the relevant mechanistic claim is no longer “a single transition near loss-curve feature *X*” but “two distinct transitions whose token-resolved timing and shape depend on data and architecture.”

10.2 What a theory of attention sinks must explain

A complete mechanistic account of attention sinks needs to satisfy at least three observed constraints from our developmental panel:

1. **The L0/L1 zero-BOS floor** (Section 5). No gradient signal installs BOS-attractor heads at L0 or L1 at any point during training, across all three models. This is a property of where in the network BOS can do useful work, not a property of the data that eventually produces sinks. A theory of sinks needs to predict that the first two layers are off-limits for sink installation.
2. **The data-dependent shape** (Section 6, comparing Pythia and OLMo). At fixed architecture (dense) and scale (1B), DCLM produces a sharper transition and higher saturation than Pile. The data does not just change the asymptote; it changes the sharpness of the transition. A theory that predicts the asymptote from data alone (e.g., from BOS-token frequency in the corpus) is insufficient.
3. **The MoE smoothing-and-reduction effect** (Section 6, comparing OLMo and OLMoE). At fixed data (DCLM) and scale (1B active parameters), the MoE architecture both reduces the

asymptotic BOS-class fraction and smooths the dynamics from a sharp transition to a gradual ramp. The architectural intervention changes two things at once. A theory should ideally identify a single underlying mechanism that manifests as both reduction and smoothing.

A natural candidate framework that captures (1) is the “residual-stream null-space” picture: at L0 the residual stream is one token’s embedding plus position; at L1 the residual stream is L0’s embedding plus the L0 attention output, which by (1) is itself content-dependent and never sink-routed. Sink heads can only do useful no-op work once the residual stream carries a token-mix from ≥ 2 prior layers of attention. We do not develop this picture quantitatively; it is the kind of theory the L0/L1 floor calls for.

10.3 Bridge to parameter-space emergence

This paper studies emergence in *activation space*: PR is a spectral statistic of per-head attention outputs over an evaluation batch. The capability selectivity ratio is an averaged attention pattern on the same batch. Both quantities are activation-side, not weight-side.

A complementary program studies the same emergence event from the parameter-side. Xu [2026d] tracks low-dimensional optimizer-induced drift and transverse dynamics through training, and reports phase-transition-like changes in the parameter-space geometry at training-token counts that coincide, in their panel, with capability-emergence events identified by downstream evaluation. Xu [2026c] examines spectral-edge dynamics of training trajectories and finds signal–noise geometry signatures at the same training-token counts. The picture both papers develop is that parameter-space and activation-space both register the emergence event, and that the parameter-space signature can be read out without constructing a task-specific evaluation.

If the two programs are studying the same event, several questions sharpen. (i) The L0/L1 floor in activation space presumably has a parameter-space counterpart: L0 and L1 attention matrices should *not* develop the spectral signature SED associates with sink-formation in later layers. (ii) The two distinct transitions (induction vs BOS-attractor) we observe should map to two distinct parameter-space transitions, with the BOS-attractor transition’s sharp DCLM-dense character visible in OLMo’s weight spectrum. (iii) The predictive 0.3–2% training-fraction recall in Section 8 should have a parameter-space analogue: SED’s spectral statistic at the same checkpoints should identify the same set of attention heads. None of these are tested here; they are the empirical questions an integrated activation-and-parameter-space theory would need to answer.

10.4 Practical methodological consequences

Three concrete recommendations follow from the developmental findings:

1. **Apply the capability-screen at intermediate checkpoints rather than only the final one.** The 67% recall threshold is reached at 0.3–2% of total training in all three models, so most of the circuit is identifiable from a checkpoint two orders of magnitude smaller in training-token count than the final checkpoint.
2. **Distinguish “induction emergence” from “attention-sink emergence” in any developmental analysis.** They are not the same event in DCLM 1B models. Conflating them confuses what is being studied.
3. **When BOS-class fraction exceeds ~70%, use the all-head capability-specific screen rather than best-class ranking.** Best-class ranking surfaces BOS heads ahead of capability heads in this regime, which is the wrong filter for capability-specific causal claims (see Section 9’s scope statement).

11 Limitations

Checkpoint resolution. The 10 log-spaced revisions per model are the published HuggingFace checkpoints; finer-grained checkpoints would catch transitions our grid currently lumps together. Pythia 1B in particular: the 6B-token co-emergence of induction with BOS-10% may be two distinct transitions at finer resolution. The OLMo 1B sharp transition between 117B (BOS-7%) and 264B (BOS-70%) similarly is the between-checkpoint gap on a $\log_2$ -spaced grid; the underlying transition could be sharper or smoother on a $\log_{10}$ -spaced grid.

Seed. Each of the three 1B models is trained once, with a single random seed; we do not have within-architecture re-pretraining at this scale (re-pretraining a 1B-class natural-text model is expensive in compute). The findings should therefore be read as “what these specific runs produced,” not as “what the (model, data) configuration produces in expectation.” At smaller scale, six seeds of TS-51M [Xu, 2026a] show within-architecture variability in which heads implement a target capability; we cannot rule out a similar trajectory-shape variability at 1B.

Evaluation batch. The capability-selectivity ratios are computed on the synthetic induction batch (random tokens, structured $A B \dots A$ pattern). Whether the same trajectory timings hold on natural-text induction-target positions is an open question. The final-checkpoint cross-validation in Xu [2026a] indicates the synthetic-batch-identified induction circuit is the circuit doing induction on natural text in OLMoE; we do not repeat that validation per-revision in this paper because it would require natural-text batches on every checkpoint.

MoE forward-pass cost. OLMoE 1B-7B’s forward pass involves top-8 expert routing across 64 experts, which makes per-revision mech-interp roughly $2\times$ more expensive than dense OLMo at the same nominal parameter count. We sampled 8 OLMoE revisions rather than 10 for this reason. Two missing revisions are below 20B tokens, which is below the earliest published HuggingFace checkpoint anyway.

Capability classes. We tracked induction, previous-token, self, BOS-attractor, duplicate-token, and local. The class taxonomy is not exhaustive; in particular, no class in this list captures composed tasks like IOI. The cross-task generalisation of the capability-screen recipe is the subject of the companion pattern-selectivity paper [Xu, 2026b].

12 Conclusion

Tracking induction, previous-token, and BOS-attractor head emergence at 10 log-spaced revisions in each of three 1B-class language models reveals that what is sometimes called “the induction phase transition” is, on DCLM data, two transitions: capability-circuit formation by 20–25B tokens, and attention-sink formation 10–20 $\times$ later. The L0/L1 zero-BOS floor holds across all 30 runs; BOS-attractor formation follows three distinct shapes across the three (data, architecture) configurations; the capability-specific screen converges to most of the final induction circuit within $\sim 1\%$ of training; and the per-head PR signal is elevated at or before capability-selectivity threshold crossings, for every final-checkpoint induction head sampled. The picture that emerges is of *two distinct training-time transitions* whose timing and shape depend on data and architecture in separable ways, with the L0/L1 zero-BOS floor as a stable architectural constraint underneath both.

References

- Stella Biderman, Hailey Schoelkopf, Quentin G Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, et al. Pythia: A suite for analyzing large language models across training and scaling. In *International Conference on Machine Learning (ICML)*, 2023.
- Arthur Conmy, Augustine N Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, et al. The pile: An 800gb dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*, 2020.

- Dirk Groeneveld, Iz Beltagy, Pete Walsh, Akshita Bhagia, Rodney Kinney, Oyvind Tafjord, Ananya Harsh Jha, Hamish Ivison, Ian Magnusson, Yizhong Wang, et al. Olmo: Accelerating the science of language models. *arXiv preprint arXiv:2402.00838*, 2024.
- Jeffrey Li, Alex Fang, Georgios Smyrnis, Maor Ivgi, Matt Jordan, Samir Gadre, Hritik Bansal, Etash Guha, Sedrick Keh, Kushal Arora, et al. Datacomp-lm: In search of the next generation of training sets for language models. *arXiv preprint arXiv:2406.11794*, 2024.
- Tom Lieberum, Matthew Rahtz, János Kramár, Neel Nanda, Geoffrey Irving, Rohin Shah, and Vladimir Mikulik. Does circuit analysis interpretability scale? evidence from multiple choice capabilities in chinchilla. *arXiv preprint arXiv:2307.09458*, 2023.
- Samuel Marks, Can Rager, Eric J Michaud, Yonatan Belinkov, David Bau, and Aaron Mueller. Sparse feature circuits: Discovering and editing interpretable causal graphs in language models. *arXiv preprint arXiv:2403.19647*, 2024.
- Niklas Muennighoff, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Jacob Morrison, Sewon Min, Weijia Shi, Pete Walsh, Oyvind Tafjord, Nathan Lambert, et al. Olmoe: Open mixture-of-experts language models. *arXiv preprint arXiv:2409.02060*, 2024.
- Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, et al. In-context learning and induction heads. *Transformer Circuits Thread*, 2022. URL <https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads/index.html>.
- Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: A circuit for indirect object identification in gpt-2 small. In *International Conference on Learning Representations (ICLR)*, 2023.
- Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations (ICLR)*, 2024.
- Yongzhong Xu. Spectral probe-circuits: A three-step recipe for identifying attention-head circuits in pretrained language models, 2026a. Companion paper (methodology).
- Yongzhong Xu. Pattern selectivity vs task-causal structure: Composed-task circuits across three 1B-class architectures, 2026b. Companion paper (cross-task generalization).
- Yongzhong Xu. Spectral edge dynamics of training trajectories: Signal–noise geometry across scales. *arXiv preprint arXiv:2603.15678*, 2026c.
- Yongzhong Xu. Optimizer-induced low-dimensional drift and transverse dynamics in transformer training. *arXiv preprint arXiv:2602.23696*, 2026d.